

Cluster #069

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1 PROJECT URL

<https://www.botto.com>

2 DESCRIPTION OF ARTWORK

Botto project is an experiment in creating a decentralized autonomous artist that generates art based on community feedback. Every week, Botto creates and sells artwork via a series of models and crowd-sourced evaluations of over 9,000 people who also decide how to manage the artist and its sales. Over the centuries, the idea of autonomous entities with creative or co-creative abilities has captivated the human imagination. From early literary examples [Kantosalo et al., 2021] to today's text-to-image models like Stable Diffusion and Midjourney, which can generate realistic imagery in seconds, the notion of machines as artistic agents has gained prominence.

Botto is the product of such dreams of creating an autonomous art agent, an ambitious experiment in autonomous art creation without direct human intervention. By embracing decentralized technologies and autonomous decision-making, Botto pushes the boundaries of artistic expression. It challenges traditional beliefs about authorship, curation, and artistic processes, forging new paths in the art world. The project leverages NFTs, DAOs, and DeFi to facilitate the sale of artworks and compensate contributors who offer feedback on Botto's images, thereby shaping its aesthetic development. However, this labor is carefully managed to preserve Botto's agency and prevent it from being a mere tool under the control of a single human artist. For instance, no human is permitted to modify Botto's images or prompts. As a pioneering community, economy, and creative force, the Botto project grapples with complex issues surrounding anthropomorphized AI, machine creativity, governance, ownership, and credit.

2.1 Approach

The Art Engine of Botto is a multifaceted system designed to generate image "fragments." The first component is a custom prompt generator that makes textual prompts by combining random words and complete sentences. These prompts serve as the inspiration

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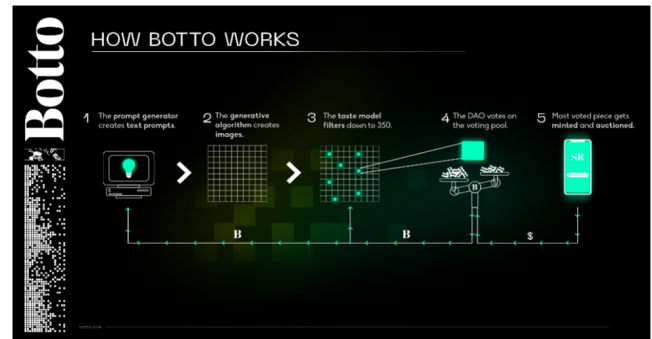


Figure 1: Diagram of Botto's process. Product of BottoDAO.

for image generation in the second stage, which involves a text-to-image model that leverages both VQGAN + CLIP [Radford et al., 2021] and Stable Diffusion to create 4,000 - 8,000 weekly images based on the provided prompts. Finally, there is a computational taste mode that filters out the best works.

The taste model presents 350 new images every week and the community members, BottoDAO, then vote to select the most aesthetically pleasing piece from this collection. This taste model and prompt generator learn from the votes and adjust their generation and selection in the subsequent week. So as to not find a niche too quickly, Botto continues to surprise and challenge the audience by seeking a number of images that have different characteristics that have not been seen before. The chosen piece from weekly voting rounds then becomes a canonical Botto artwork, minted and auctioned as an NFT through the digital auction house SuperRare. The proceeds go back into the community treasury, half of which is directly paid out to the voters and the other half managed to pay for costs like servers and exhibitions.

The original concept and design of Botto's art engine were conceived by Mario Klingemann (aka Quasimondo), while the studio Eleven Yellow and crypto investment company Carbono have been actively involved in the ongoing design process and operation of the BottoDAO. By embracing the idea of an autonomous artistic agent detached from human creators, Botto aligns with the notion of the machine condition [Colton et al., 2020], wherein machines can creatively express their own existence. At the same time, Botto makes the generative process into an ongoing dialogue between the artists, participants, and the artwork itself, allowing for continual exploration, iteration, and refinement.

2.2 Case Study: Cluster #069

Botto was conceived as an entity that would evolve with the advancements in technology. Adding capabilities would enable it to grow with the times, though would need to be implemented with



Figure 2: Cluster #069. Product of BottoDAO, shown on Feral File. <https://feralfile.com/artworks/71b22ba1-02b9-4994-a903-27bda5073fda>

care so as to be able to run autonomously and not break its agency as an artist. Cluster:#069 is the result of a commission from the digital art platform Feral File looking to feature digital artists' processes in its show "Simulation Sketchbook: Works in Process". For the work, a clustering algorithm was applied to the entirety of Botto's body of work to organize the autonomous artist's fragments into common artistic themes. This revealed 128 themes within Botto's works. For each of these themes, 64 fragments were chosen from the categories and clustered by the algorithm based on common features. This process resulted in a sort of style swatch, creating a mosaic from the fragments. Style grids were created for each of the 128 themes, upon which Botto's community voted to select the most popular piece. This process was the final step in the production of Cluster:#069.

Botto's use of its own fragments to create a composite image calls attention to the underlying mechanical functions at work and their dependence upon human labor. Shedding light on the themes of social influence, the data labor economy, and collective intelligence, Cluster:#069 examines the boundaries of artist and process. The symbiotic and reflexive relationship between art and technology throughout time is showcased in this project, as popular aesthetic themes are defamiliarized and altered by the perception of many authors as opposed to a singular one. In addition, it pushes both participants and viewers to reflect on whether art is the image or the process. And if it's a process, how much of the art is the instructions vs. the environment? In this new art form, Botto as a whole (the DAO, the art engine, the voting process) is the art of Botto as much as the images which it produces. This offers some freedom from old mythology around how we define ownership

in art. Not only that but old antiquated structures built to value art become re-contextualized, as these new mediums which are seemingly infinitely reproducible usher in paradigm shifts around scarcity and creativity, as well as the value of collective authorship. Martin Lukas Ostachowski has said of crypto-based art that "it is a movement of values over aesthetics. As such it is essential to remember and promote the concept of self-sovereign crypto artists" [Francheschet, et al., 2021] Blockchain technology created tools for artists to challenge their role in the traditional art world.

3 ARTIST INFORMATION

Botto is a decentralized autonomous artist, initially conceptualized by Mario Klingemann, and governed by a collective of stakeholders through a DAO (decentralized autonomous organization). Its artistic capabilities are based on a custom art engine, a proprietary blend of algorithmic models that have synthesized millions of works of art. Mario Klingemann is an artist who uses algorithms and artificial intelligence to create and investigate systems. He has been recognized as a pioneer in the field of AI art, neural networks and machine learning. He has worked with prestigious institutions including The British Library, Cardiff University and New York Public Library, and was Artist in Residence at Google Arts and Culture. His artworks have been shown at MoMA New York, the Metropolitan Museum of Art New York, the Photographers' Gallery London, ZKM Karlsruhe, and Centre Pompidou Paris. Klingemann received the British Library Labs Artistic Award 2016 and in 2018 won the Lumen Prize Gold Award, which celebrates artworks made with technology, and was awarded with the Honorary Mention in the Prix Ars Electronica 2020. His installation *Memories of Passersby I* made history in March 2019 as the first autonomous AI machine to be successfully auctioned at Sotheby's. Simon Hudson is part of the core team behind Botto, his role includes operating BottoDAO and designing the governance system, economy, and research and contributor programs. Prior to joining Botto, Simon worked as a writer and researcher at a private AI research lab, Element AI, in Montreal where he investigated subjects in AI governance and human-machine collaboration. In addition to his work on Botto he also does research in science communications for AI literacy and interface design. Julia Pryde Thompson is an artist and researcher who works at the intersection of art and technology. Her work explores how we learn and share information, how perception shapes understanding, and ways we can create collectively. She is a core team member of BottoDAO, and facilitates as well as produces Botto's collaborations with other artists and art institutions. Prior to working with Botto, she was a member of Refik Anadol's studio, researching how machine learning applied to archival information might better help us comprehend the complexities of our multidimensional histories and world.